URBAN PULSE

Smart City Mobility Intelligence Platform for Bengaluru

Team D

Abstract

Urban Pulse is a Smart City Analytics platform engineered using Microsoft Power BI to consolidate multiple, independently maintained urban mobility datasets pertaining to the city of Bengaluru into a single, coherent analytical environment. The platform draws upon data from the Bengaluru Metropolitan Transport Corporation (BMTC), municipal traffic sensors, meteorological records, road-accident registries maintained by the Bengaluru Traffic Police, ward-level demographic surveys, and ride-hailing operations data from Namma Yatri. These heterogeneous sources are cleansed, standardized, and reorganized according to a star-schema data warehouse design, which enables high-performance analytical querying and simplifies the construction of interactive dashboards.

The primary contribution of this project, undertaken in the role of Data Engineer, is the design and implementation of a scalable, maintainable dimensional model, the formulation of a comprehensive set of Data Analysis Expressions (DAX) measures, and the definition of Key Performance Indicators (KPIs) — including two original composite indices, the Mobility Access Index and the City Mobility Health Score — that allow city administrators to evaluate the performance of Bengaluru's transportation ecosystem holistically. This report documents the requirements, planning methodology, data engineering pipeline, analytical model, and visualization strategy adopted throughout the project lifecycle.

The data engineering workflow adopted for this project followed a disciplined, phase-wise approach beginning with the acquisition of raw, heterogeneous source files and culminating in a validated, query-ready dimensional model. Considerable emphasis was placed on data quality: missing values, duplicate records, inconsistent data types, and non-uniform naming conventions present in the original BMTC, traffic, weather, demographic, accident, and ride-hailing datasets were systematically identified and resolved using Power Query prior to the construction of fact and dimension tables. This rigorous transformation process ensured that the resulting star-schema warehouse could support accurate, high-performance analytical queries across all seven dashboard pages of the final report.

Beyond its technical contribution, Urban Pulse is intended to serve as a practical demonstration of how self-service business-intelligence tools can be applied to real-world civic data challenges. By consolidating six previously siloed data domains into a single, coherent analytical narrative, the project illustrates the value of dimensional modelling and well-governed DAX measure design in transforming raw operational data into insight that is directly usable by non-technical decision-makers. The methodology, data model, and analytical framework presented in this report are additionally structured to be extensible, so that future work — including real-time data ingestion and predictive analytics, as discussed in the Future Scope section — can be layered onto the existing foundation with minimal architectural rework.

Keywords: Smart City Analytics, Power BI, Star Schema, Data Warehousing, DAX, Urban Mobility, Bengaluru, Key Performance Indicators.

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1. Introduction

Rapid urbanisation across Indian metropolitan regions has placed sustained pressure on transportation infrastructure, public transit systems, and traffic management authorities. Bengaluru, widely recognised as India's leading technology hub, exemplifies this challenge: its road network services a population exceeding one crore residents, a dense public bus network operated by the Bengaluru Metropolitan Transport Corporation (BMTC), an expanding ride-hailing ecosystem, and a road-safety landscape that continues to demand active monitoring. Despite the availability of granular operational data from each of these domains, the data itself is fragmented across disparate systems, formats, and custodians, which prevents any single stakeholder from forming a comprehensive, city-wide understanding of mobility performance.

Urban Pulse was conceived to close this gap. The project integrates six independent Bengaluru-specific datasets — public transit (GTFS), road traffic, weather, demographics, road-accident records, and ride-hailing activity — into a unified Power BI analytical solution. Rather than presenting each dataset in isolation, Urban Pulse organises the underlying data according to a dimensional (star-schema) model, which permits cross-domain analysis: for example, correlating rainfall intensity with traffic congestion, or relating ward-level population density to ride-hailing demand. The resulting platform is intended to function as a decision-support tool for city planners, transport authorities, and researchers who require a consolidated, interactive view of Bengaluru's mobility ecosystem.

This report documents the complete engineering process undertaken to build Urban Pulse, from requirement gathering through to dashboard visualization, with particular emphasis on the data engineering and analytical modelling responsibilities carried out during the project.

2. Problem Statement

Urban mobility data relevant to Bengaluru is generated and maintained by a variety of independent organisations, including the BMTC, traffic-monitoring authorities, the India Meteorological Department, the Bengaluru Traffic Police, census and demographic survey agencies, and private ride-hailing operators such as Namma Yatri. Each of these organisations records data using its own schema, granularity, update frequency, and file format. Consequently, no single authority or system currently offers a consolidated view that links public transit performance, traffic conditions, weather patterns, road-safety outcomes, population characteristics, and ride-hailing demand within one analytical environment.

This fragmentation has direct operational consequences. City administrators evaluating the effectiveness of public transit investments cannot easily determine whether poor transit reliability correlates with adverse weather or high-traffic corridors. Road-safety officials cannot readily assess whether accident hotspots align with areas of high ride-hailing or pedestrian density. Urban planners lack a single reference point from which to prioritise infrastructure investment across wards. Urban Pulse directly addresses this problem by constructing a unified data warehouse and an accompanying suite of interactive dashboards that allow these cross-domain relationships to be explored efficiently and reliably.

3. Project Objectives

The specific objectives defined for the Urban Pulse project are as follows:

1. To identify, acquire, and consolidate publicly available Bengaluru mobility datasets spanning public transit, traffic, weather, demographics, road safety, and ride-hailing domains.
2. To design and implement a normalised, query-efficient star-schema data warehouse capable of supporting multi-domain analytical queries.
3. To perform systematic data cleaning, transformation, and standardisation using Power Query, ensuring consistency of data types, formats, and naming conventions across all source datasets.
4. To define a comprehensive library of DAX measures that quantify transit reliability, traffic congestion, ride-hailing performance, road safety, and demographic characteristics.
5. To design original, composite Key Performance Indicators — namely the Mobility Access Index and the City Mobility Health Score — that synthesise multiple operational metrics into executive-level indicators.
6. To develop a suite of interactive Power BI dashboards that enable drill-down analysis, cross-filtering, and slicer-based exploration for each analytical domain.
7. To document the complete engineering methodology in a manner suitable for academic evaluation and future extension.

4. Scope of the Project

The scope of Urban Pulse is confined to the descriptive and diagnostic analytics of Bengaluru's mobility ecosystem using historical and semi-static datasets. The platform does not perform real-time data ingestion, nor does it provide predictive or prescriptive analytics such as traffic forecasting or route optimisation; these are identified as candidate extensions in the Future Scope section of this report. The project scope explicitly includes:

- Data engineering activities: acquisition, cleaning, transformation, and modelling of six mobility-related datasets.
- Design of a multi-fact star-schema warehouse suitable for Power BI consumption.
- Definition and implementation of DAX-based analytical measures and KPIs.
- Construction of seven dashboard pages covering executive summary, traffic, transit, weather, road safety, demographics, and ride-hailing domains.

Excluded from scope are real-time streaming ingestion pipelines, machine-learning-based forecasting models, mobile application development, and integration with live governmental control systems. These exclusions were made in consideration of the academic timeframe and the available data sources, which are static exports rather than live feeds.

5. Requirement Analysis

Requirement analysis for Urban Pulse was carried out in two categories, functional and non-functional, in accordance with standard software engineering practice. Functional requirements describe the specific capabilities the system must provide, while non-functional requirements describe the quality attributes the system must exhibit.

5.1 Functional Requirements

The system shall satisfy the following functional requirements:

8. Import multiple heterogeneous Bengaluru mobility datasets from their respective source formats (CSV, GTFS text files, and tabular exports).
9. Clean and standardise the imported data, including the resolution of missing values, duplicate records, inconsistent data types, and non-uniform column naming.
10. Construct a star-schema data model comprising clearly separated fact and dimension tables.
11. Establish validated relationships between fact and dimension tables to support accurate cross-table aggregation.
12. Generate a library of reusable DAX measures covering transit, traffic, weather, road safety, demographic, and ride-hailing analytics.
13. Provide interactive dashboards with drill-down functionality, cross-filtering, and slicer-based exploration.
14. Present composite and elementary KPIs suitable for city-wide mobility monitoring by non-technical stakeholders.

5.2 Non-Functional Requirements

In addition to the functional capabilities listed above, the system is required to satisfy the following quality attributes:

Attribute	Description
Performance	The star-schema design must ensure sub-second query response for standard dashboard interactions.
Scalability	The data model must accommodate the addition of new fact or dimension tables without structural redesign.
Usability	Dashboards must be navigable by non-technical stakeholders through intuitive slicers and visuals.
Interactivity	Cross-filtering between visuals must be supported across all dashboard pages.
Maintainability	DAX measures must be reusable, centrally organised, and clearly named.
Reliability	Data transformation logic must be repeatable and produce consistent results on refresh.

Table 5.1 — Non-functional requirements of the Urban Pulse platform

6. Data Sources

Six independent datasets were identified, acquired, and integrated into the Urban Pulse data warehouse. Each dataset was selected on the basis of its relevance to a specific dimension of urban mobility and its availability in a structured, analysable format. The datasets and their respective analytical purposes are summarised in Table 6.1.

Dataset	Purpose
BMTC GTFS	Public transit route, stop, and schedule analysis
Traffic Dataset	Traffic congestion and average speed analysis
Weather Dataset	Assessment of weather impact on mobility patterns
Bengaluru Demographics	Ward-level population and literacy analysis
BTP Accident Dataset	Road-safety and accident-severity analysis
Namma Yatri Ride-Hailing Dataset	Ride-hailing demand and conversion analysis

Table 6.1 — Source datasets integrated into Urban Pulse

The General Transit Feed Specification (GTFS) dataset published by BMTC provided the foundation for public-transit analytics, including route definitions, stop locations, agency information, and scheduled trip data. The traffic dataset supplied interval-based measurements of vehicular volume, average speed, and congestion levels across monitored corridors. Weather records provided daily temperature and rainfall observations. The demographic dataset, sourced at ward level, provided population counts disaggregated by gender along with literacy statistics. The Bengaluru Traffic Police accident dataset provided station-wise counts of fatal and non-fatal accidents. Finally, the Namma Yatri dataset provided ward-level ride-hailing metrics, including search volume, booking volume, completion rate, cancellation rate, driver earnings, and trip distance.

7. Technology Stack

The technology stack adopted for Urban Pulse was selected to balance analytical capability with accessibility, given the academic context of the project and the availability of Microsoft Power BI as a self-service business-intelligence platform. Table 7.1 summarises the tools used at each layer of the solution.

Layer	Technology
Business Intelligence Tool	Microsoft Power BI Desktop
Data Cleaning and Transformation	Power Query (M Language)
Data Modelling	Star Schema (dimensional modelling)
Analytical Language	Data Analysis Expressions (DAX)
Version Control	GitHub
Documentation	Microsoft Word

Table 7.1 — Technology stack adopted for the Urban Pulse platform

Power BI Desktop was selected as the primary development environment on account of its integrated support for data import, transformation, modelling, and visualization within a single tool. Power Query was used for all extract-transform-load (ETL) operations, allowing data-cleaning logic to be captured as a repeatable, auditable sequence of applied steps. DAX was used to implement all derived measures and KPIs. GitHub was adopted for version control of documentation and supporting scripts, ensuring that the project history remained traceable throughout development.

8. Data Warehouse Design

Urban Pulse adopts a multi-star schema architecture, in which several fact tables — each corresponding to a distinct operational domain — share a common set of conformed dimension tables wherever a meaningful relationship exists. This design was selected in preference to a single-fact, fully normalised (snowflake or third-normal-form) model because star schemas are optimised for the read-heavy, aggregation-intensive query patterns that characterise Power BI dashboards, and because they simplify the DAX formulas required to compute cross-domain measures.

8.1 Fact Tables

Fact tables store the quantitative, event-level or measurement-level data captured from each source system. The following fact tables were implemented:

- Fact_Trips — records of individual BMTC transit trips.
- Fact_StopTimes — scheduled and actual stop-level arrival and departure events.
- Fact_Traffic — interval-based traffic volume, speed, and congestion measurements.
- Fact_Demographics — ward-level population and literacy figures.
- Fact_RideHailing_Ward — ward-level ride-hailing search, booking, and completion metrics.
- BTP_Station_Wise — station-wise accident counts, disaggregated by severity.

8.2 Dimension Tables

Dimension tables store descriptive attributes used to filter, group, and label the measures computed from the fact tables. The following dimension tables were implemented:

- Dim_Date and Dim_Calendar — calendar attributes supporting time-intelligence analysis.
- Dim_Route, Dim_Stop, Dim_Agency, and Dim_Shapes — GTFS-derived transit dimensions.
- Dim_District and Dim_Ward — administrative and geographic groupings used across demographic, ride-hailing, and accident analysis.
- Dim_Geography — spatial reference attributes supporting map-based visuals.
- Dim_Weather — daily weather observation attributes.

Relationships between fact and dimension tables were established as one-to-many relationships with single-directional cross-filtering in the majority of cases, in accordance with star-schema best practice, in order to avoid ambiguous filter propagation and circular dependency errors within the Power BI model.

9. Project Planning and Methodology

The project was executed in five sequential phases, each with clearly defined deliverables. This phased methodology allowed data-engineering activities to be completed and validated before analytical and visualization work commenced, thereby reducing rework caused by downstream discovery of data-quality issues.

9.1 Phase 1 — Requirement Gathering

The initial phase focused on identifying the datasets available for Bengaluru's mobility ecosystem, studying the structure of the General Transit Feed Specification, and evaluating candidate business-intelligence tools. Power BI Desktop was selected on the basis of its integrated ETL, modelling, and visualization capabilities.

9.2 Phase 2 — Data Collection

The second phase involved the systematic collection of the six datasets described in Section 6: BMTC GTFS, traffic, weather, demographics, accident, and ride-hailing data. Each dataset was reviewed for completeness, structural consistency, and licensing suitability prior to inclusion in the project.

9.3 Phase 3 — Data Engineering

The third phase, and the primary focus of the Data Engineer role within this project, involved data cleaning, missing-value handling, duplicate-record removal, data-type conversion, column-name standardisation, relationship creation, and star-schema design. Each source dataset was processed independently in Power Query prior to being loaded into the unified data model, ensuring that transformation logic remained modular and independently testable.

9.4 Phase 4 — Analytics

The fourth phase involved the design of DAX measures, the definition of KPIs, and the planning of dashboard layouts. This phase translated the cleaned dimensional model into the analytical vocabulary — reusable measures and composite indicators — required by the visualization layer.

9.5 Phase 5 — Visualization

The final phase involved the construction of seven dashboard pages: Executive, Traffic, BMTC, Weather, Road Safety, Demographics, and Ride Hailing. At the time of writing, this phase remains in progress, with dashboard layout, visual selection, and cross-page navigation being iteratively refined.

Phase	Focus Area	Status
1	Requirement Gathering	Completed
2	Data Collection	Completed
3	Data Engineering	Completed
4	Analytics (KPIs & DAX)	Completed
5	Visualization (Dashboards)	In Progress

Table 9.1 — Summary of project phases and current status

10. Key Performance Indicators

A central analytical contribution of Urban Pulse is its library of Key Performance Indicators, which range from elementary, single-domain metrics to composite, multi-domain indices designed specifically for this project. KPIs were selected to provide city administrators with both granular operational insight and executive-level summary indicators.

10.1 Mobility Access Index (Custom KPI)

The Mobility Access Index is an original composite KPI designed to measure the accessibility of formal mobility services — encompassing both public transit and ride-hailing — relative to the size of the population served. It is computed as:

```
Mobility Access Index =
(Total BMTC Trips + Completed Ride-Hailing Trips) / Total Population
```

A higher value of the Mobility Access Index indicates greater accessibility to mobility services on a per-capita basis, and can be used to compare relative service provision across wards or districts.

10.2 City Mobility Health Score (Custom KPI)

The City Mobility Health Score is a second original composite KPI intended to provide a single, executive-level indicator of overall mobility performance across the city. It is calculated as a weighted combination of four sub-components — traffic performance, transit performance, ride-hailing performance, and road safety — expressed on a 0–100 scale:

```
City Mobility Health Score =
  0.40 x Traffic Performance
+ 0.30 x Transit Performance
+ 0.20 x Ride-Hailing Performance
+ 0.10 x Road Safety
```

Each sub-component is itself derived from underlying operational measures, as summarised in Table 10.1.

Sub-Component	Formula	Weight
Traffic Performance	Average Speed / Maximum Speed	40%
Transit Performance	Completed Trips / Planned Trips	30%
Ride-Hailing Performance	Completed Bookings / Searches	20%
Road Safety	1 – (Fatal Accidents / Total Accidents)	10%

Table 10.1 — Composition of the City Mobility Health Score

10.3 Domain-Specific KPIs

In addition to the two composite indices described above, a set of domain-specific KPIs was defined to support detailed, drill-down analysis within each of the seven dashboard pages. These are summarised in Table 10.2.

KPI	Formula	Domain
Transit Reliability	Completed Trips / Scheduled Trips	Public Transit

KPI	Formula	Domain
Transit Demand	Total Trips or Total Stop Visits	Public Transit
Ride-Hailing Demand	Total Searches	Ride Hailing
Ride-Hailing Conversion Rate	Bookings / Searches	Ride Hailing
Ride Completion Rate	Completed Trips / Bookings	Ride Hailing
Cancellation Rate	Cancelled Trips / Bookings	Ride Hailing
Driver Earnings	Sum of Driver Earnings	Ride Hailing
Average Fare	Average Fare per Trip	Ride Hailing
Average Trip Distance	Average Distance per Trip	Ride Hailing
Traffic Congestion Index	Traffic Volume / Average Speed	Traffic
Road Safety Index	1 – (Fatal Accidents / Total Accidents)	Road Safety
Population Density	Population / Area	Demographics
Literacy Rate	Average Literacy Rate	Demographics

Table 10.2 — Domain-specific Key Performance Indicators

It should be noted that the Mobility Access Index and the City Mobility Health Score are project-specific composite indicators without a recognised industry-standard formula. The weighting scheme adopted for the City Mobility Health Score — 40% traffic, 30% transit, 20% ride-hailing, and 10% road safety — reflects an analytical design decision made for this project, based on the relative contribution of each domain to overall city mobility, and should be documented and interpreted as a custom metric rather than as an established benchmark.

11. DAX Measures Implementation

All analytical measures used within Urban Pulse were implemented as centrally organised DAX measures held within a dedicated Measure_Table, in accordance with Power BI modelling best practice. Organising measures in this manner improves maintainability, avoids duplication of business logic across visuals, and allows measures to be reused consistently across all seven dashboard pages. The core measures implemented for each analytical domain are presented below.

11.1 Public Transit (BMTC) Measures

```
Total Trips = COUNTROWS(Fact_Trips)

Total Stops = COUNTROWS(Dim_Stop)

Total Routes = COUNTROWS(Dim_Route)

Total Agencies = COUNTROWS(Dim_Agency)
```

11.2 Traffic Measures

```
Average Speed = AVERAGE(Fact_Traffic[Average Speed])
```

```
Traffic Volume = SUM(Fact_Traffic[Traffic Volume])

Average Congestion = AVERAGE(Fact_Traffic[Congestion Level])
```

11.3 Weather Measures

```
Average Temperature = AVERAGE(Dim_Weather[Temperature])

Average Rainfall = AVERAGE(Dim_Weather[Rainfall])
```

11.4 Road Safety Measures

```
Total Accidents = SUM(BTP_Station_Wise[Total Accidents])

Fatal Accidents = SUM(BTP_Station_Wise[Fatal Accidents])

Non Fatal Accidents = SUM(BTP_Station_Wise[Non Fatal Accidents])
```

11.5 Demographic Measures

```
Total Population = SUM(Fact_Demographics[Population])

Male Population = SUM(Fact_Demographics[Male])

Female Population = SUM(Fact_Demographics[Female])

Literacy Rate = AVERAGE(Fact_Demographics[Literacy Rate])
```

11.6 Ride-Hailing Measures

```
Total Searches = SUM(Fact_RideHailing_Ward[Searches])

Total Bookings = SUM(Fact_RideHailing_Ward[Bookings])

Completed Trips = SUM(Fact_RideHailing_Ward[Completed Trips])

Cancelled Trips = SUM(Fact_RideHailing_Ward[Cancelled Bookings])

Driver Earnings = SUM(Fact_RideHailing_Ward[Drivers' Earnings])

Average Fare = AVERAGE(Fact_RideHailing_Ward[Average Fare per Trip])

Average Distance = AVERAGE(Fact_RideHailing_Ward[Average Distance per Trip (km)])
```

These elementary measures serve as building blocks for the composite KPIs described in Section 10. For example, the Traffic Congestion Index is computed as the ratio of the Traffic Volume measure to the Average Speed measure, while the Road Safety Index reuses the Fatal Accidents and Total Accidents measures defined above. This layered approach — elementary measures composed into higher-order KPIs — ensures that any correction to an underlying calculation automatically propagates to all dependent indicators.

12. Dashboard Design

The Urban Pulse report is organised into seven dashboard pages, each corresponding to a distinct analytical domain, in addition to a consolidated executive view. Every dashboard page provides slicers for common dimensions such as date, ward, and district, KPI cards summarising headline metrics, and a combination of chart types selected for their suitability to the underlying data.

Dashboard Page	Primary Content
Executive Dashboard	City Mobility Health Score, Mobility Access Index, and cross-domain summary visuals
Traffic Dashboard	Average speed, traffic volume, and congestion index by corridor and time
BMTC Dashboard	Trip counts, route coverage, and transit reliability by agency and route
Weather Dashboard	Temperature and rainfall trends and correlation with mobility metrics
Road Safety Dashboard	Accident counts, fatality ratios, and station-wise safety index
Demographics Dashboard	Population distribution, gender split, and literacy rate by ward
Ride Hailing Dashboard	Search, booking, completion, cancellation, and earnings metrics by ward

Table 12.1 — Dashboard pages and their primary analytical content

Each dashboard supports slicer-based filtering, cross-visual highlighting, and drill-down navigation from ward or district level down to individual record detail, enabling users to move fluidly between a city-wide overview and localised, granular analysis.

13. Expected Outcomes and Results

Upon completion, Urban Pulse is expected to deliver the following outcomes:

- A unified, queryable view of Bengaluru's urban mobility ecosystem spanning six previously disconnected data domains.
- Data-driven insight to support transport-planning decisions at the ward and district level.
- Identification of traffic congestion hotspots through the Traffic Congestion Index and associated visuals.
- Continuous monitoring of public-transit reliability through the Transit Reliability measure.
- Quantified assessment of ride-hailing demand, conversion, and cancellation patterns.
- Analysis of the relationship between weather conditions and mobility performance.
- Systematic monitoring of road safety through accident analytics and the Road Safety Index.
- Demographic insight to support equitable, population-aware mobility planning.
- A single executive dashboard summarising city-wide mobility performance through the City Mobility Health Score.

14. Conclusion

Urban Pulse demonstrates that a rigorous, dimensionally modelled data-engineering approach can successfully unify heterogeneous, independently maintained urban datasets into a single analytical platform capable of supporting city-wide decision-making. Through the application of a multi-star schema, systematic data-cleaning procedures, and a comprehensive library of DAX-based measures and KPIs, the project translates raw operational data from six distinct domains into actionable insight regarding public transit reliability, traffic congestion, weather impact, road safety, demographic distribution, and ride-hailing performance. The two original composite indicators developed for this project — the Mobility Access Index and the City Mobility Health Score — illustrate how domain-specific metrics can be synthesised into executive-level indicators suitable for non-technical stakeholders.

The project fulfils its stated objectives with respect to requirement analysis, data engineering, and analytical-model design, while the visualization phase remains in progress at the time of writing. The resulting platform provides a solid technical foundation upon which further analytical and predictive capability may be developed.

15. Future Scope

Several avenues exist for extending Urban Pulse beyond its current scope:

15. Integration of real-time or near-real-time data feeds from traffic sensors and BMTC vehicle-tracking systems to support live monitoring rather than static, historical analysis.
16. Incorporation of predictive analytics, such as short-term traffic-congestion forecasting or ride-hailing demand prediction, using machine-learning models trained on the existing data warehouse.
17. Extension of the geographic dimension to support finer-grained spatial analysis, including geospatial heat-mapping of accident and congestion hotspots.
18. Development of a mobile-accessible version of the dashboard suite to support field-based decision-making by transport authorities.
19. Expansion of the data warehouse to incorporate additional mobility modes, such as metro ridership and non-motorised transport infrastructure.

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